

Global CO₂ Emissions:

Who Emits, How Much, and What's Changing

Fossil Fuels & Industry · 1750 – 2024 · Policy & Research Edition

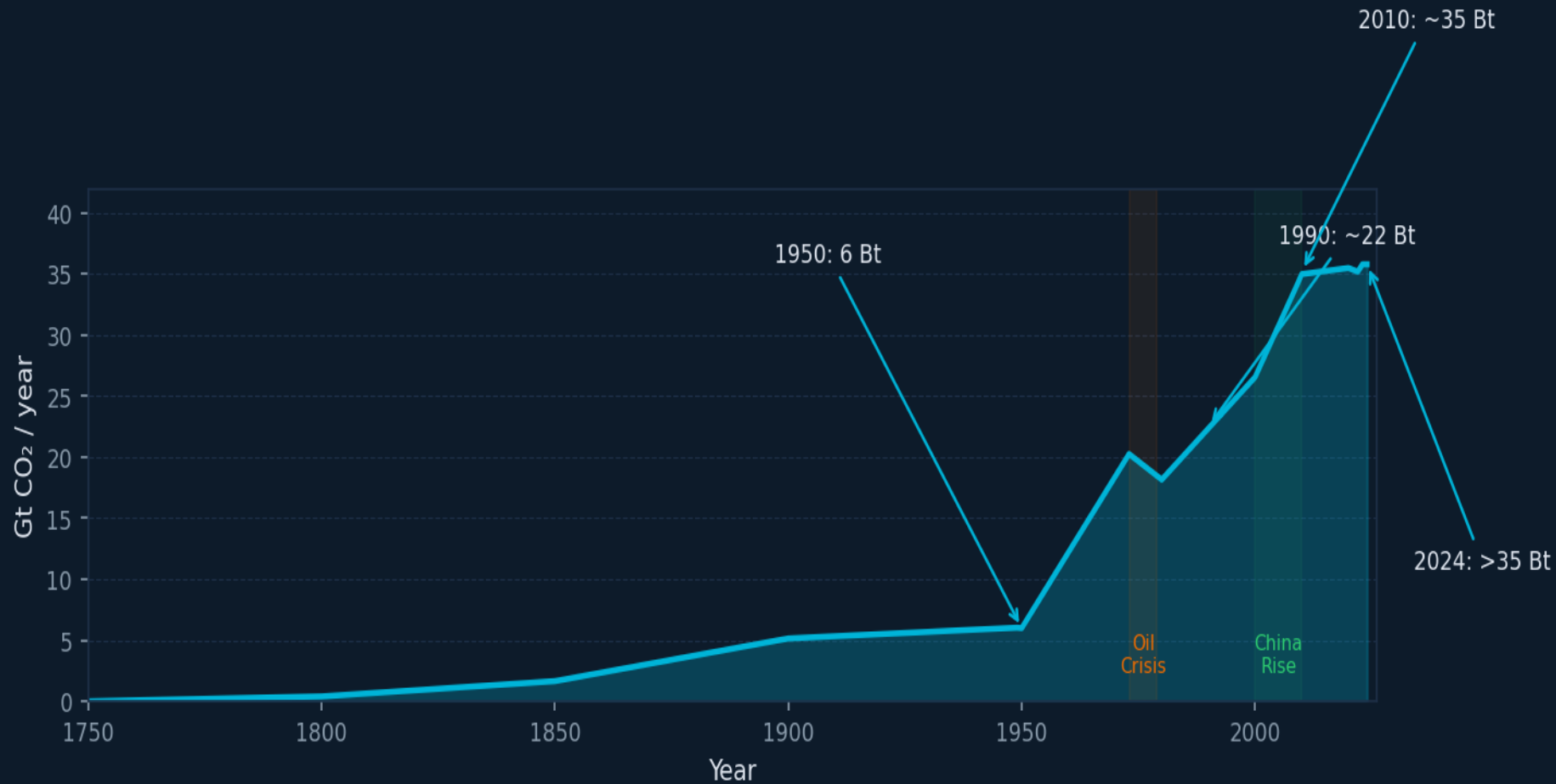
Data: Global Carbon Budget 2025 · via Our World in Data

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For: Policy Analysts · Climate Negotiators · Sustainability Researchers

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Fossil-fuel & industry CO₂ | 1750–2024 | Gt CO₂/yr

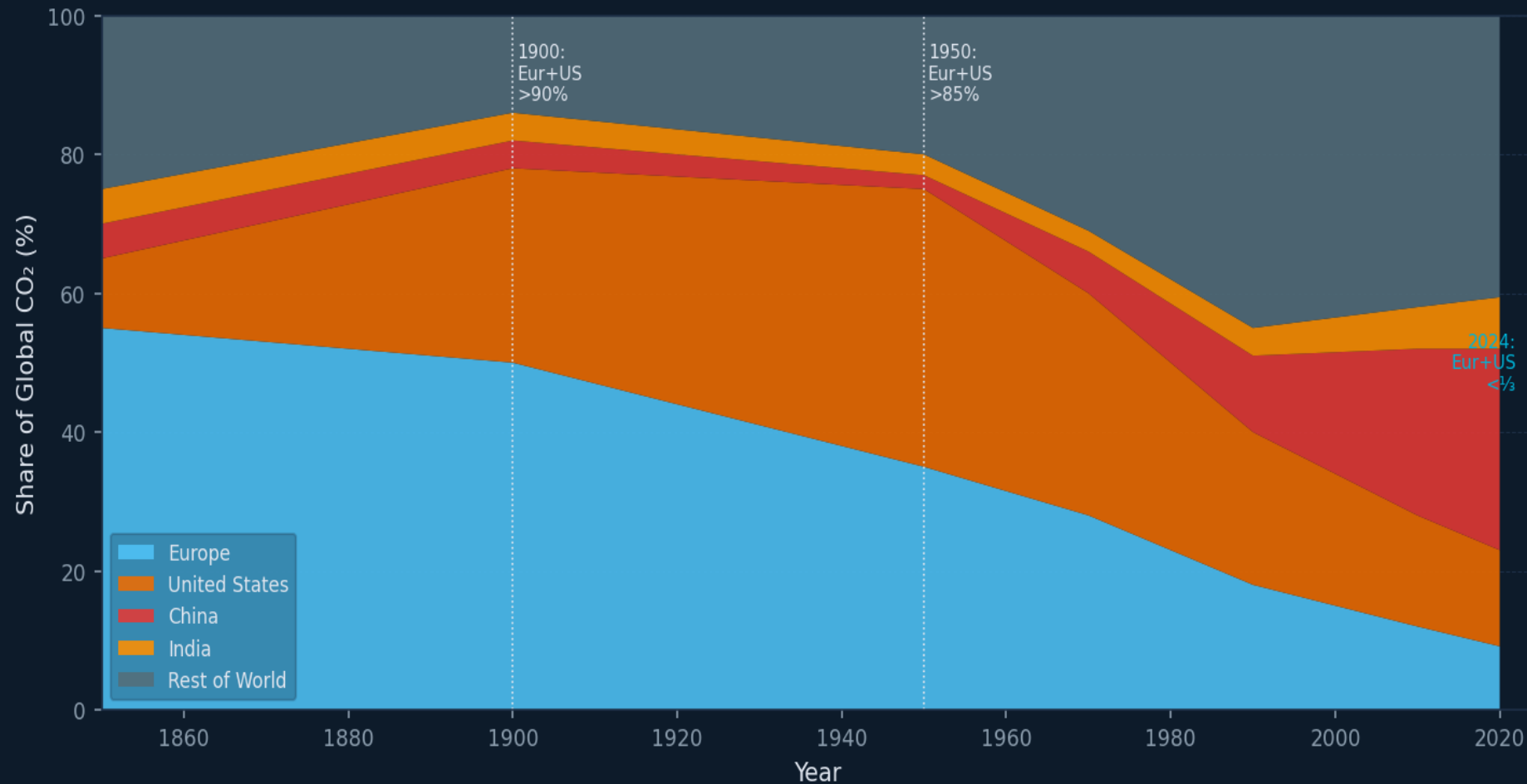


Key Inflection Points

- ▶ Pre-industrial era: near-zero
- ▶ Post-WWII surge (1950–73)
- ▶ Oil shock pause (1973–79)
- ▶ China-led boom (2000–10)
- ▶ Growth slowing — but **NO peak yet**

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

Share of global CO₂ emissions by region | 1850–2024



The Historic Inversion

1900 Eur + US > 90%

1950 Eur + US > 85%

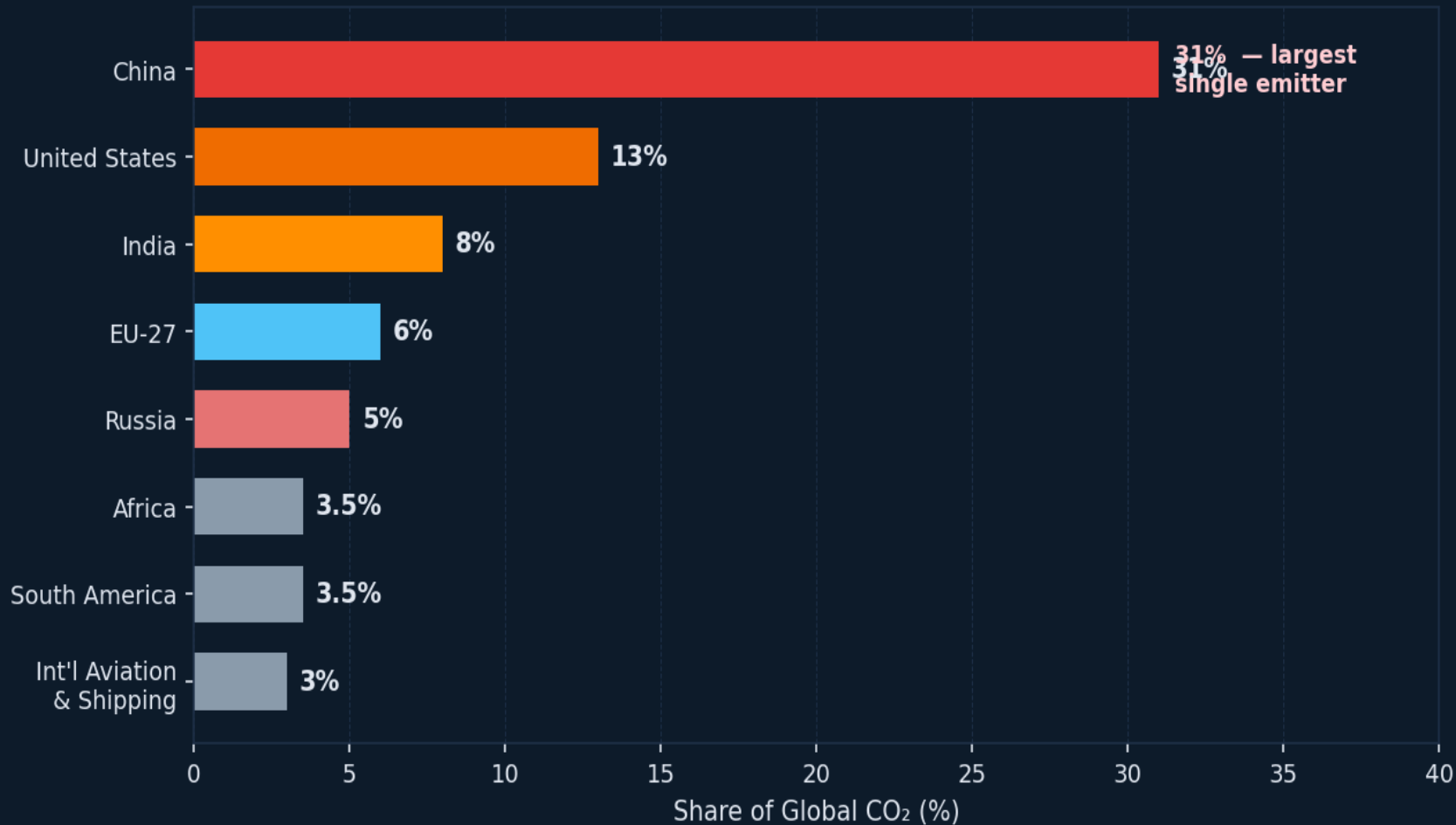
2024 Eur + US < 33%

Asia now produces

~50% of global CO₂

China Now Emits More Than One-Quarter of Global CO₂

Share of global CO₂ emissions by country/region | 2024



2024 Snapshot

China ~31% Rising, slowing

United States ~13% Declining

India ~8% Rising

EU-27 ~6% Declining

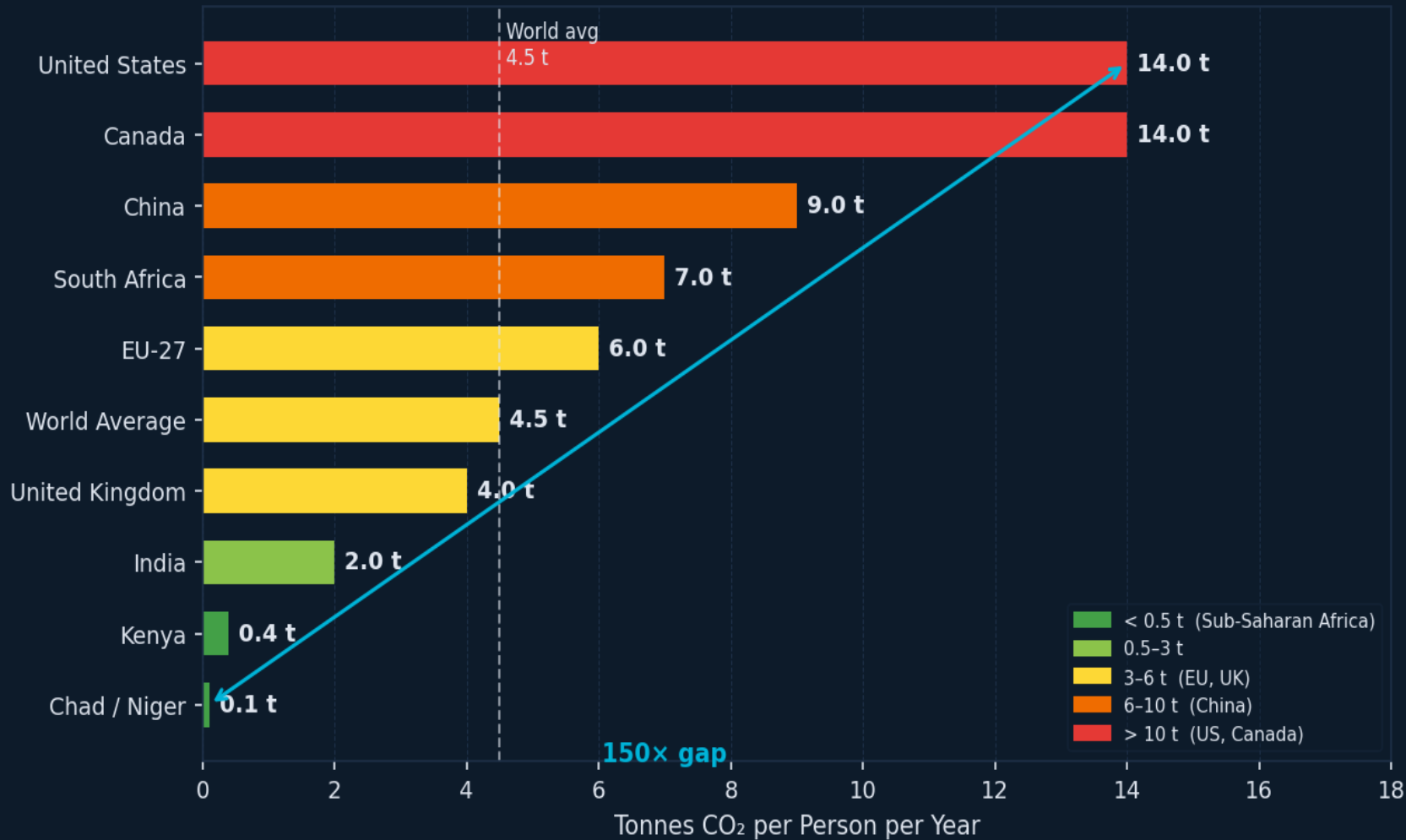
Russia ~5% Stable

Africa ~3–4% Slowly rising

S. America ~3–4% Stable

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

CO₂ per person (tonnes) | approx. 2024 | 150x gap highlighted



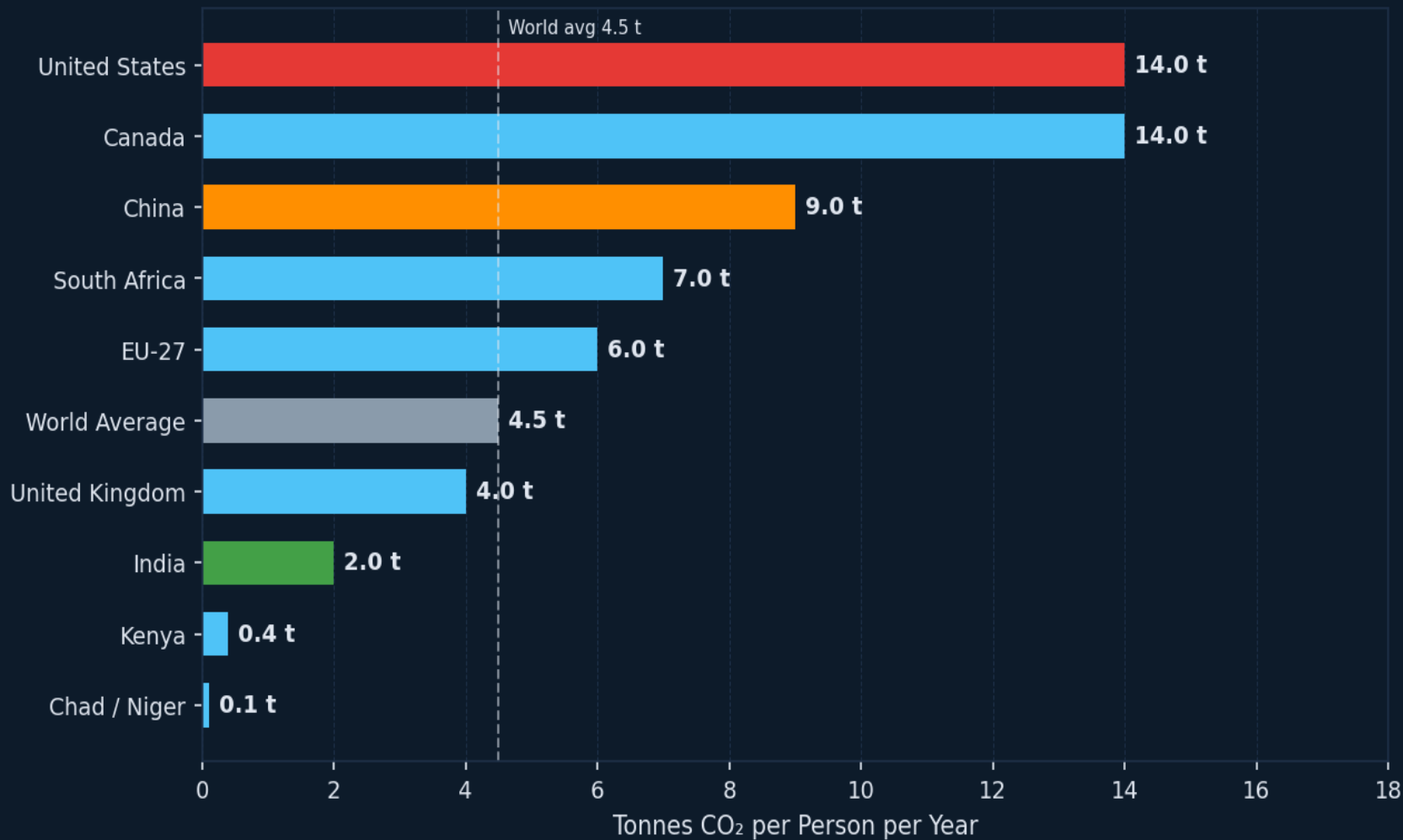
The Inequality Story

The average American emits in under 2 days what the average person in Mali emits in a year.

France & UK (~4 t) emit far less than the US (~14 t) despite similar living standards.

Per Capita Emissions: ~14 t in the US vs. ~9 t in China and ~2 t in India

CO₂ per person (tonnes) | approx. 2024 | key country comparison



Key Observations

China is the #1 total emitter — but per capita (9 t) remains well below the US (14 t).

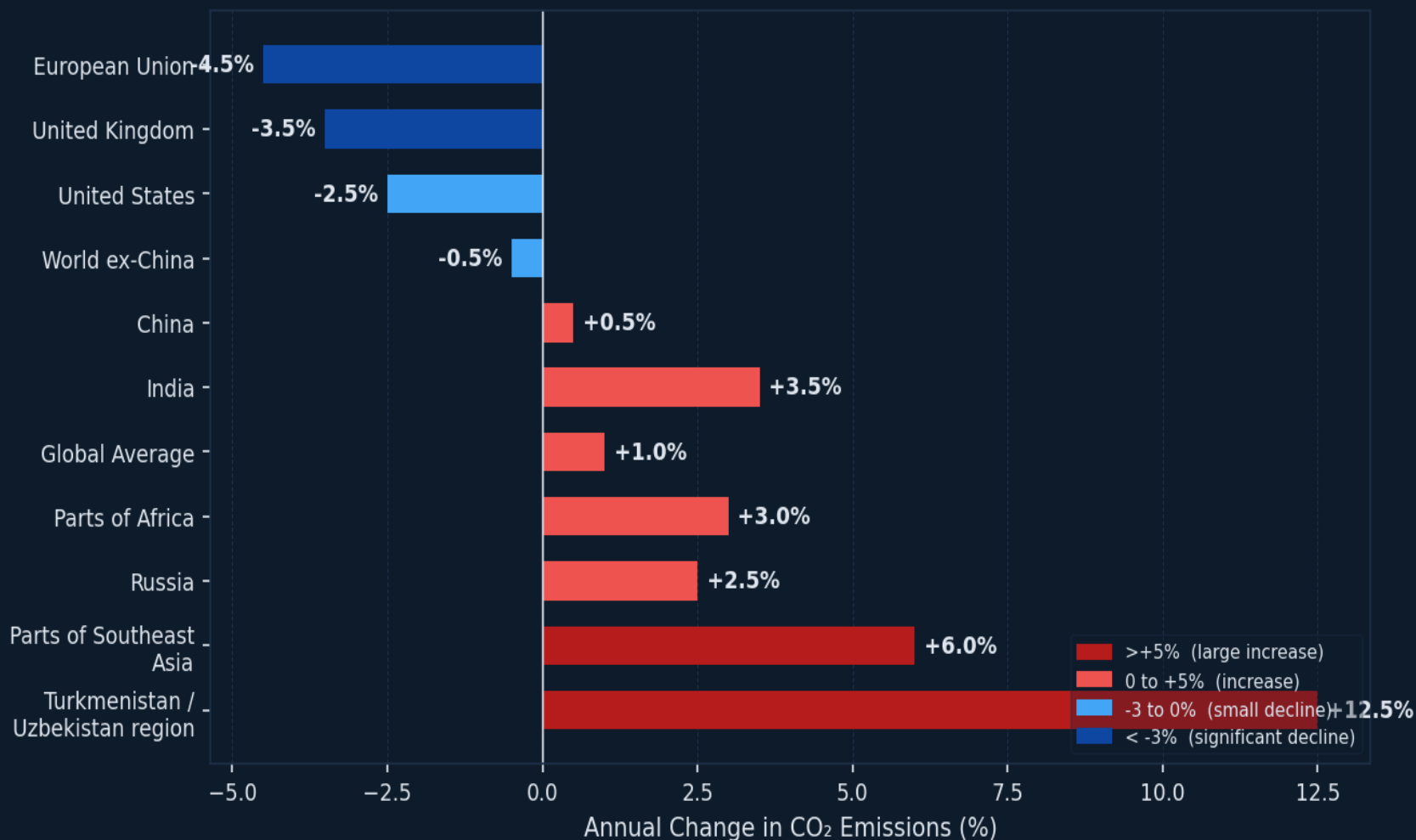
India at 2 t/person: still far below the world average of 4.5 t.

UK (4 t) $\approx$ World average — enabled largely by nuclear & renewables.

See *Global Carbon Budget 2025 / Our World in Data (figure_8)* for historical per-capita trend lines.

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

Annual % change in CO₂ emissions | 2024 | Diverging bar chart



2024 Signal

Blue (declining): US, EU, UK — driven by renewable growth & efficiency.

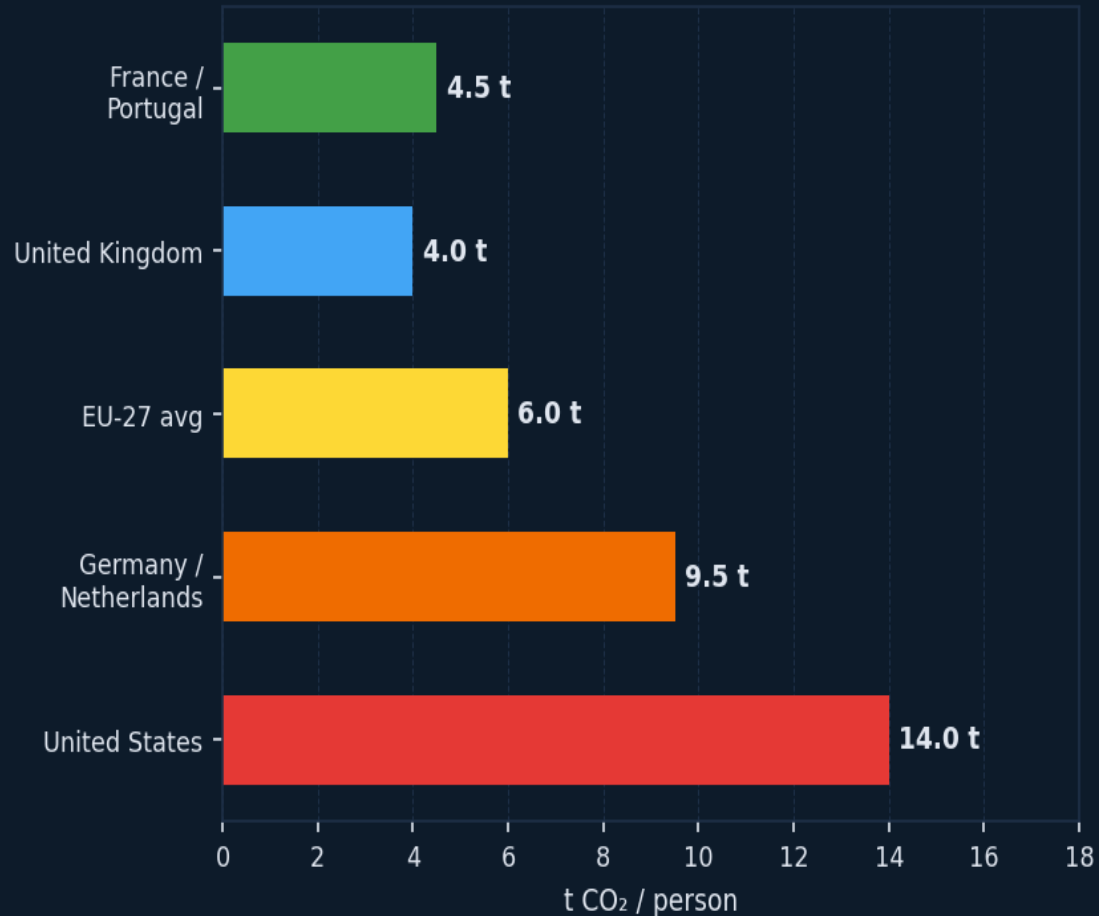
Red (rising): Russia, SE Asia, Africa, Central Asia (+10%).

Net global result: emissions still above 35 Bt — no peak yet.

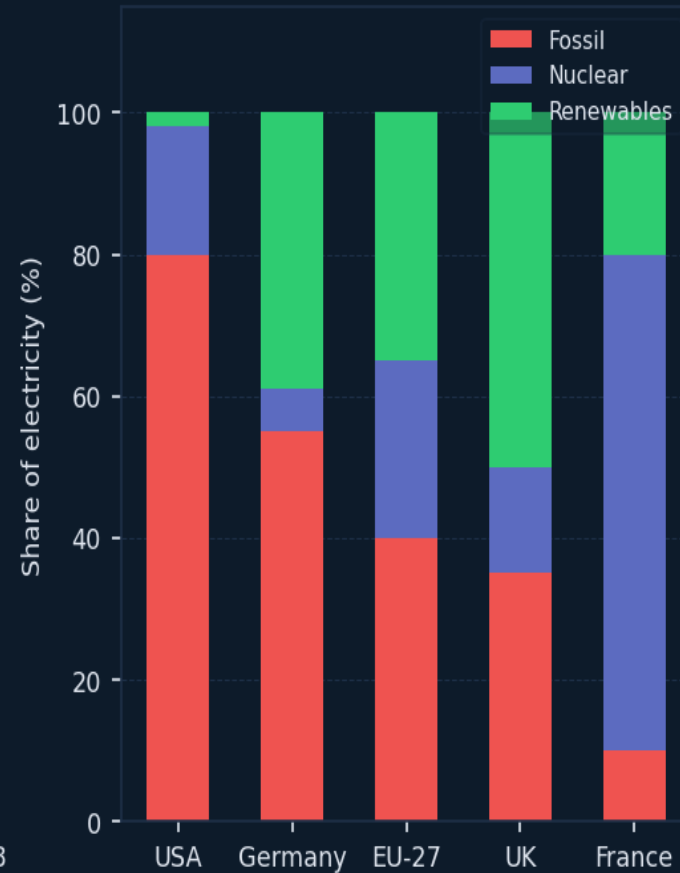
Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Per capita CO₂ (t/person, ~2024) · Qualitative energy mix comparison

Per Capita CO₂



Electricity Mix (approx.)



Policy Insight

Similar GDP, very different emissions — the gap is almost entirely explained by electricity source.

France: ~80% nuclear → ~4.5 t/person

UK: ~50% low-carbon → ~4 t/person

US: ~80% fossil → ~14 t/person

Note: electricity mix figures are approximate and illustrative.

Key Takeaways: Responsibility Is Shared but Unequal

Global Carbon Budget 2025 · Our World in Data · Fossil Fuels & Industry 1750–2024

01 No peak yet

Global CO₂ from fossil fuels exceeds 35 Bt/yr and continues to rise — the world has not yet reached peak emissions despite slowing growth.

02 China is the largest annual emitter — but not per capita

China emits ~31% of global CO₂, but per capita (9 t) remains well below the US and Canada (~14 t). Total vs. per-capita framing changes the responsibility narrative.

03 Historical responsibility skews toward the West

Europe and the US dominated emissions throughout the 20th century (>90% in 1900, >85% in 1950). The cumulative atmospheric stock of CO₂ reflects this legacy.

04 150× per capita gap persists

Sub-Saharan Africa emits ~0.1 t/person while the US and Canada emit ~14 t — a 150× disparity. The lowest emitters face the highest climate vulnerability.

05 Energy policy can decouple prosperity from emissions

France, the UK, and Portugal show that high living standards are achievable at 4–5 t/person through nuclear and renewable energy. Policy and technology choices are decisive.